GE Medical Systems

Technical Publications

Direction 2209610
Revision 0

SignaR Horizont
Base to HiSpeed with SGD Upgrade

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785–5052/8*323–5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section "S" of the Policy & Procedure Bulletins.

3/12/92

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A' CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Rev 0	Mar 11, 1998	Initial Release

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	0								
Damage in Trans.	–								
Direction 2128126	0*								
A	0								
i	0								
1–1 to 1–5	0								
2–1 to 2–13	0								
3–1	0								

* This revision number/letter corresponds to the indicated document’s revision control system.

TABLE OF CONTENTS

SECTION 1 – OVERVIEW

1–1 SIGNA HORIZON SYSTEM CONFIGURATION

This manual provides instructions to upgrade a Signa Horizon Base System to Signa Horizon with SGD HiSpeed System. To install this upgrade on a Horizon 5.x system, Release 5.7 (or later) software is required. This installation manual is applicable to Fixed–Sites at magnet field strengths of 1.0T or 1.5T.

This Direction covers the following catalogs:

- M1090WD – Horizon Base to HiSpeed SGD Upgrade

Note

There are some naming differences between the sales literature documents and the technical documentation the catalog in this section. Listed below are the corresponding names. The Pre–Installation document uses the technical document names.

SALES LITERATURE NAME	=	TECHNICAL DOCUMENT NAME
SX Gradient Subsystem	=	SGD (Scaleable Gradient Driver)
Signa Horizon	=	Signa Horizon with SGD or Signa Horizon 5.x with SGD
Signa Horizon LX	=	Signa Horizon LX with SGD or Signa Horizon 8.x with SGD

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2142460, Signa Horizon Upgrade Pre–installation,* Pre–installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon system are listed in *Direction 2142460, Signa Horizon Upgrade Pre–installation*, Section 10, TOOLS AND TEST EQUIPMENT.

1–3 PRODUCT LOCATOR

Product Locator System tracks creation, shipment, trans–shipment, and installation location of serialized models.

The installation card is not a "warranty card", therefore it is to be processed as soon as the installation starts.

The PLC system also includes Remanufacture, Scrap, and Transfer cards. They should be obtained and completed for all serialized cabinets removed from the system according to Policy and Procedure.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. Process just the installation card and discard any extra shipment cards and labels.

System upgrade serial–number rating plates are furnished. They provide the means to track systems with various hardware versions of the System Upgrade. Kit added rating plates are furnished with various upgrade kits. They signify that model numbers on the major component have changed function and compatibility and have become a "look–alike" for current production in performance and compatibility. These rating plates do not have shipment cards since serial numbers are not assigned.

1–4 DISCARDED MATERIAL RETURN POLICY

The "discard" or "return pile" consists of items removed by the Signa Horizon Upgrade or traded in for new options and have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the return pile is per GE Medical Systems Policy and Procedures.

1. For coordination of return pile, contact GEMS Recycling Center: Phone (414)–747–6997 or Dial Comm 8*579–6997; Fax (414–747–6855); WizMail "RECYCLING"; Field ASPEN 1–800–525–1516, Box #26041.

When calling, the following information is needed:

- FDO (Future Delivery Order) number under which the upgrade was shipped
- Site name, address, and phone number
- Date of completed Horizon upgrade.
- Name and Beeper Number of Field Engineer that could be available when the equipment is picked up.

2. Prepare the return pile for shipment as follows:
3. Process product locater cards for all serialized components removed from the system according to policy.
4. Regulations governing the disposal of this material in the recycle pile vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the "recycle pile" per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1–5 UPGRADE SEQUENCE

Follow the Horizon system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that the upgrade has arrived complete and site is ready to shut down.
- Shut site down.
- Install HiSpeed upgrade
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Functional Checks
- System calibrations
- System performance checks
- Cover replacement
- Complete final tasks and return site to customer.

1–6 SYSTEM POWER SHUTDOWN

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance HighSpeed line measurements have been obtained and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.

DANGER!!

ELECTRIC SHOCK HAZARD!! HIGH VOLTAGES ARE PRESENT ON INTERCONNECTING CABLES USED WITH SPECIAL FIELD INSTALLED ION–PUMPS USED WITH OXFORD MAGNETS. THE ION PUMP POWER SUPPLY IS USUALLY CONNECTED TO A POWER SOURCE THAT IS NOT CONTROLLED BY THE PDU AND HIGH VOLTAGE WILL BE PRESENT EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK, DISCONNECT POWER TO THE ION PUMP POWER SUPPLY, LOCK OUT, AND TAG BEFORE PERFORMING PROCEDURES IN THE FOLLOWING SECTIONS. ALSO CHECK FOR OTHER INTERCONNECTING EQUIPMENT WHICH WAS FIELD INSTALLED AND DISCONNECT POWER, LOCK OUT, AND TAG.

4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
7. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring.

1–7 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD–ROM, Set–Up & Calibration, and perform the following:
   - PDU Voltage Checks Compact
   - Startup Checks Compact
   - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the SGD Gradient Cabinet power modules, switch all cabinet circuit breakers to ON. The SGD power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.

1–8 FINAL COORDINATION

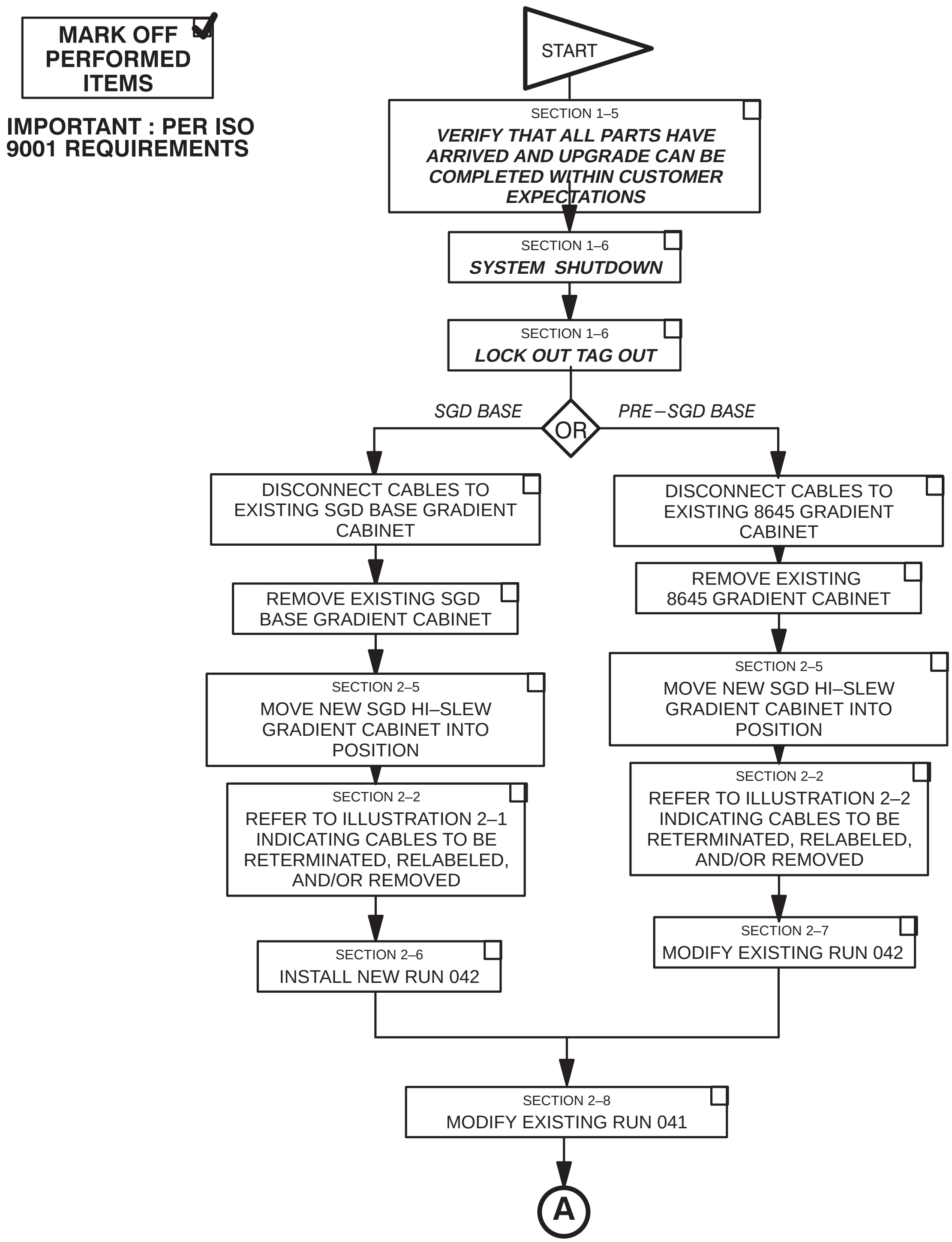
1. Replace cabinet covers that have not been previously replaced.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locater installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

Product Locator File
P.O. Box 414, W–523
Milwaukee, WI 53201–0414

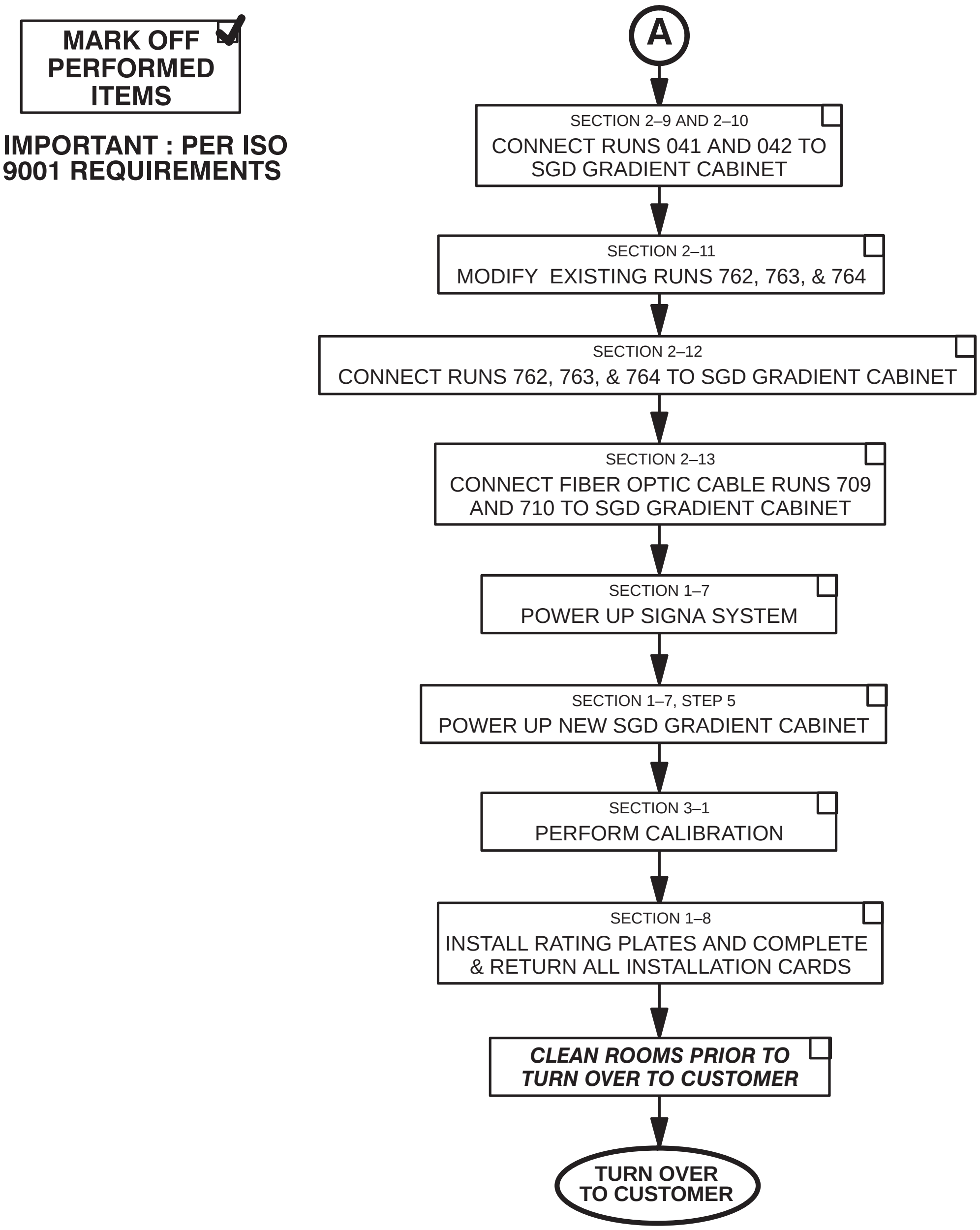
Note
Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI’s.

SECTION 2 – MECHANICAL UPGRADE PROCESS

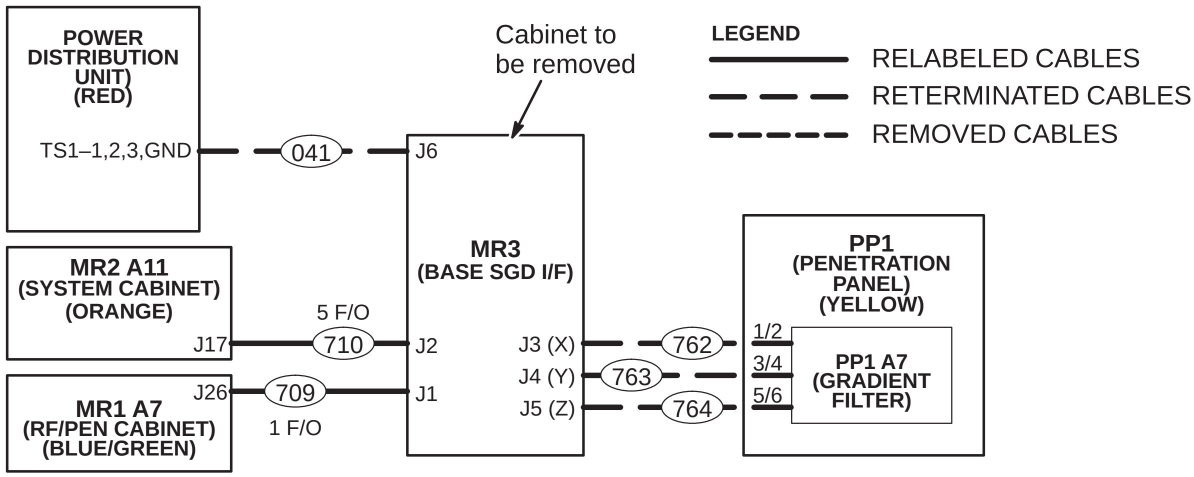
2–1 INSTALLATION FLOWCHART



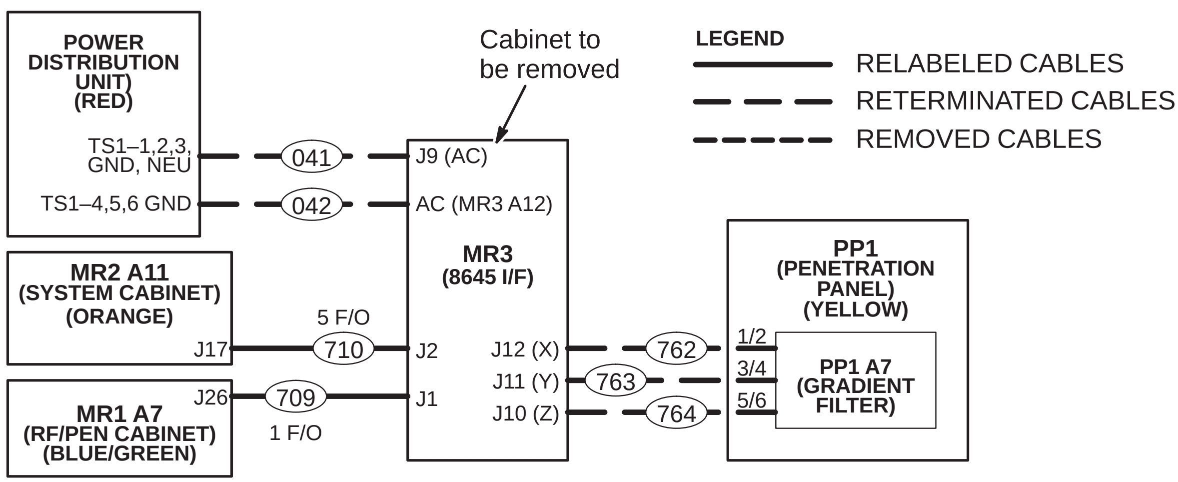
2–1 INSTALLATION FLOWCHART (cont'd)



2–2 CABLE MAPS FOR CABLES TO BE REMOVED, RETERMINATED, AND/OR RELABELED



AFFECTED SIGNA HORIZON™ CABLES WHEN UPGRADING FROM 1.5T SGD BASE TO HiSpeed with SGD
ILLUSTRATION 2–1



AFFECTED SIGNA HORIZON™ CABLES WHEN UPGRADING FROM 1.5T BASE TO HiSpeed with SGD
ILLUSTRATION 2–2

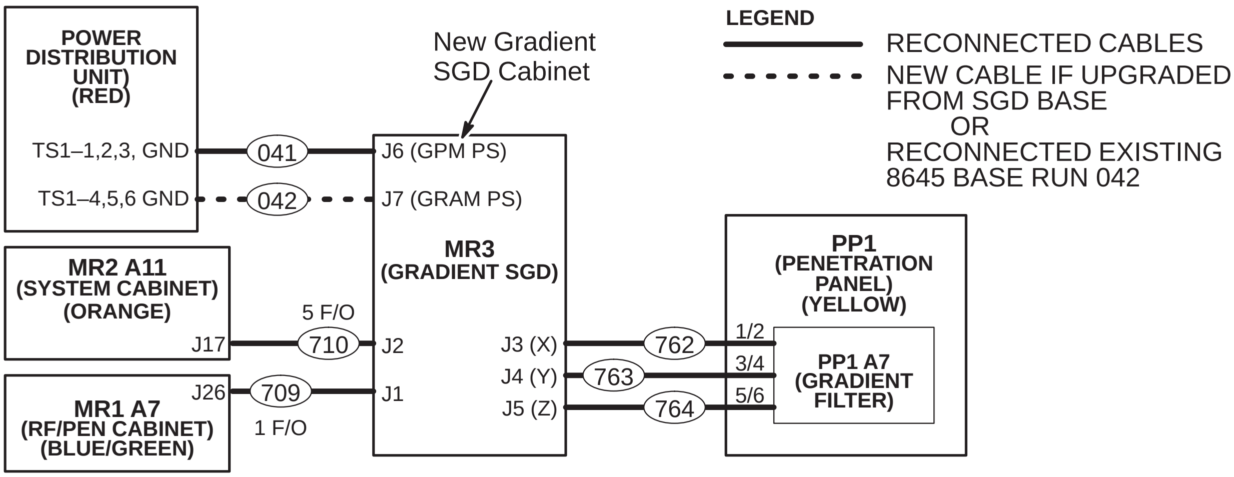
2–3 EXISTING CABLES TO BE RELABELED

Several existing Horizon 8.x cables need to be re–labeled and then re–connected per the new labels. Refer to Table 2–1.

TABLE 2–1
HORIZON 8.x CABLES RE–LABELED AND RE–LOCATED FOR UPGRADE TO HORIZON 8.x WITH SGD

OLD RUN	ONE END OLD LABEL	ONE END NEW LABEL	OTHER END
041	RUN #041 MR3–A11–J9 46–271363G3	RUN #041 MR3CABI/FJ6	NO CHANGE
042	RUN #042T MR3–A12 46–271363G604	RUN #042 MR3CABI/FJ7	NO CHANGE
709	RUN #709 MR3–A11–J1 46–317077G802	RUN #709 MR3CABI/FJ2	NO CHANGE
709–1	RUN #709–1	RUN #709–1 MR3GIPJ6	NO CHANGE
710	RUN #710TR–1 MR3–A11–J2 46–317085G602	RUN #710TR–1 MR3CABI/FJ1	NO CHANGE
710–1	RUN #710–1	RUN #710–1 MR3GIPJ2	NO CHANGE
710–2	RUN #710–2	RUN #7710–2 MR3GIPJ3	NO CHANGE
710–3	RUN #710–3	RUN #710–3 MR3GIPJ4	NO CHANGE
710–4	RUN #710–4	RUN #710–4 MR3GIPJ1	NO CHANGE
710–5	RUN #710–5	RUN #710–5 MR3GIPJ5	NO CHANGE
762	RUN #762 MR3–A11–J12 2100948–4	RUN #762 MR3CABI/FJ3	NO CHANGE
763	RUN #763 MR3–A11–J11 2100948–5	RUN #763 MR3CABI/FJ4	NO CHANGE
764	RUN #764 MR3–A11–J10 2100948–6	RUN #764 MR3CABI/FJ5	NO CHANGE

2–4 INTERCONNECT MAP AFTER UPGRADE



SIGNA HORIZON™ CABLES AFTER UPGRADING TO HiSpeed with SGD
ILLUSTRATION 2–3

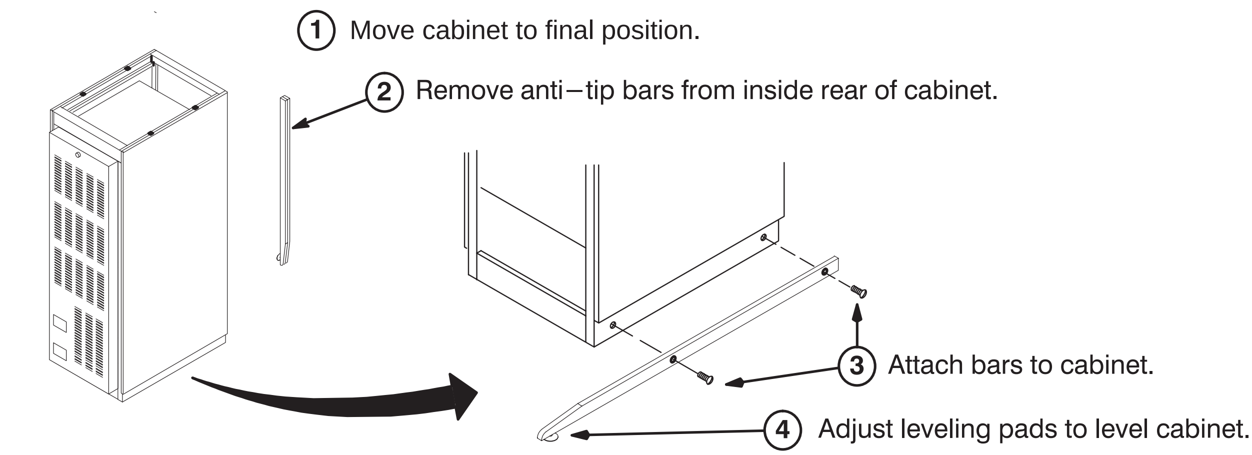
DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

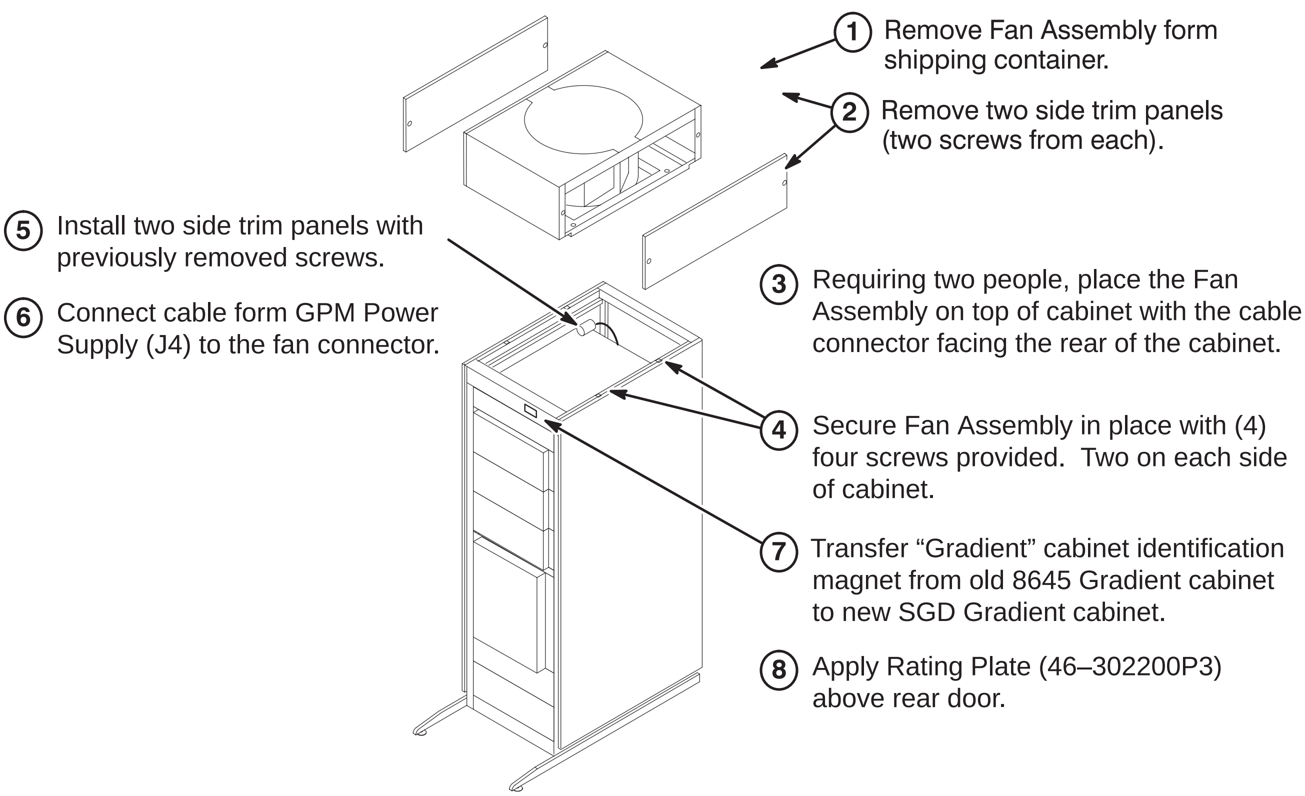
2–5 NEW SGD HI–SLEW GRADIENT CABINET

1. Verify that Runs 762, 763, and 764 have sufficient length to route from SGD Hi–Slew Cabinet to Penetration Panel.
2. Verify that Runs 041 and 042 have sufficient length to route from SGD Hi–Slew Cabinet to PDU.

2–5–1 Position SGD Cabinet



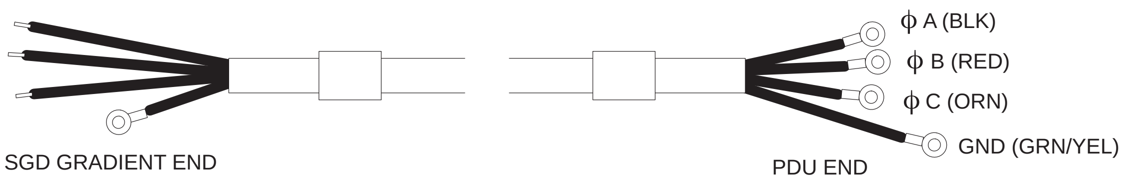
2–5–2 Install Fan Assembly and Attach Insite Identification Magnet



2–6 NEW RUN 042 POWER CABLE

2–6–1 Power Cable Installation Planning

1. Verify that power cables, Runs 041 and 042, are present. A new Run 042 (46–271363G6) will be required if upgrading from SGD Base. If so, perform the following steps.



NEW RUN 042, 46–271363G6

- **If sufficient room exists to store excess cable length**:
  Plan for storage if cable is not cut to length. Then proceed to procedure **2–6–3** step 3, and **2–6–4**.
- **If insufficient room exists to store excess cable length**:
  Cable is cut shorter and re–terminated. Complete procedures **2–6–2, 2–6–3** and **2–6–4**.

2–6–2 Power Cable Installation

1. Allow sufficient slack for routing and the cutting of cables to length between PDU and SGD cabinet.
2. Select end to be shortened. If PDU, cut and trim insulation for connection to Compact PDU cabinet TB10 pressure connectors. Two–bay PDU has needs ring terminals so it may be easier to shorten SGD Gradient end.
3. Transfer color code tape, run–number labels, and applicable wire marking labels as each cable is cut to size.

2–6–3 Power Cable Pre–connect Checks

Checks for opens, shorts, and crossed connections are made for re–terminated power cables prior to connection to PDU and SGD cabinet. This does not provide ground resistance checks for cabinets. Cabinet ground resistance and leakage current checks are found in the MR CD–ROM*,* Set Up and Calibration: System, GROUNDING AND LEAKAGE CURRENT CHECKS.

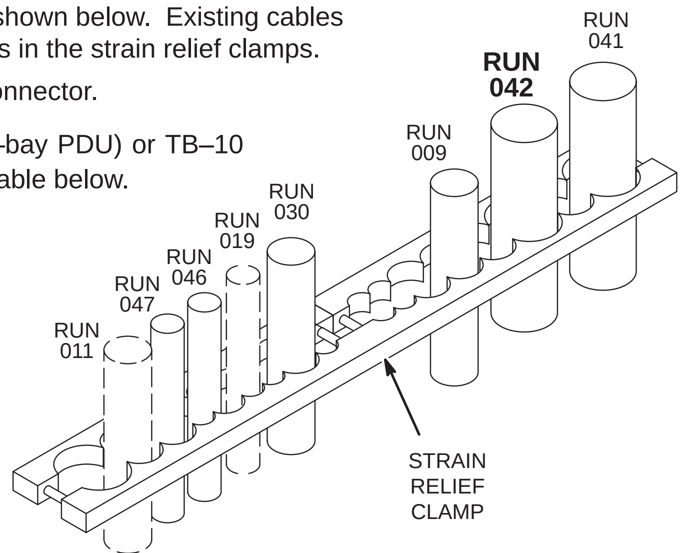
1. Use the ohmmeter and check each new power cable for the following:
   - Open circuit between cable neutral and cable ground.
   - Open circuit between cable neutral and each line on the cable.
   - Open circuit between cable ground and each line on the cable.
   - Less than 0.1 ohm between neutral and the light blue (or white) wire.
   - Less than 0.1 ohm between ground and the green/yellow wire.
2. Any out–of–spec conditions must be immediately corrected. After power cable has been satisfactorily checked out, proceed to step 3.
3. Unroll cable to lay without twists or kinks from SGD Cabinet to PDU through provided troughs, conduit, or ducts.

2–6–4 Connect Run 042 To PDU

1. Route new SGD Gradient power cable to: TS1 terminal strip (2–bay PDU) or TB–10 pressure connectors (Compact PDU)
2. Secure cables in place within strain relief clamp in PDU and tighten screws.

   Note: Compact PDU strain relief aranagement as shown below. Existing cables may have to be relocated to fit within the new cables in the strain relief clamps.
3. Trim ground wire cable and connect to ground bus connector.
4. Connect new power cable to TS1 terminal strip (2–bay PDU) or TB–10 pressure connectors (Compact PDU) as shown in Table below.

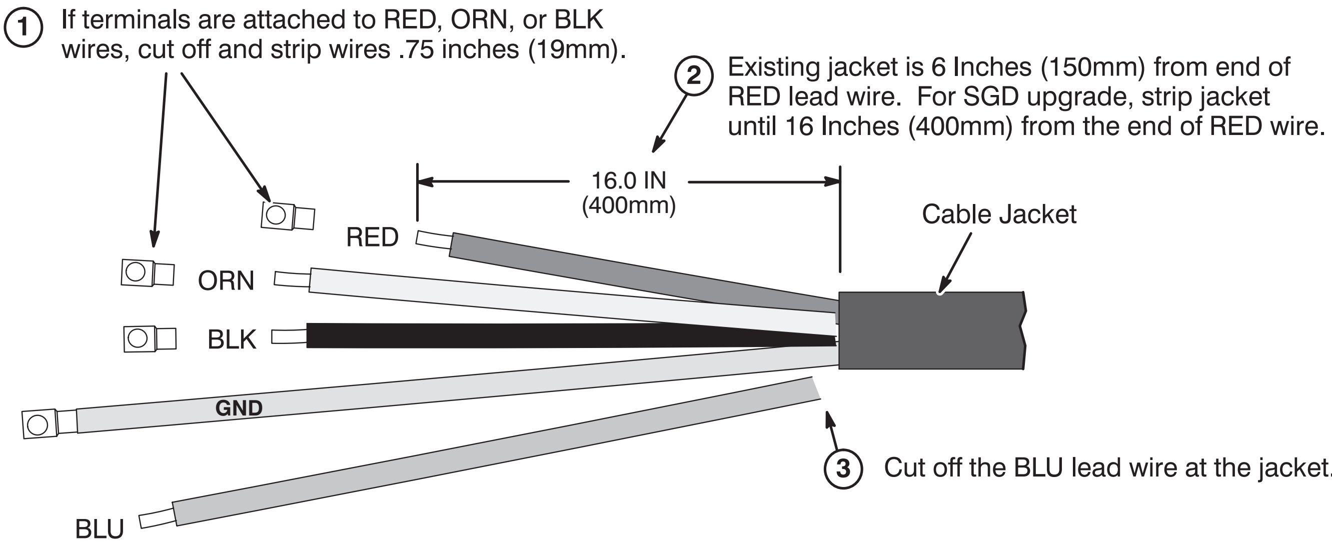
TS1/ TB10	RUN	PHASE	DESTINATION
1,2,3	041	ϕ A, B, C	SGD Gadient Cabinet 3ϕ, Ground
4,5,6	042	ϕ A, B, C	SGD Gradient Cabinet 3ϕ, Ground
COLOR CODE for cables 3ϕ = Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel			



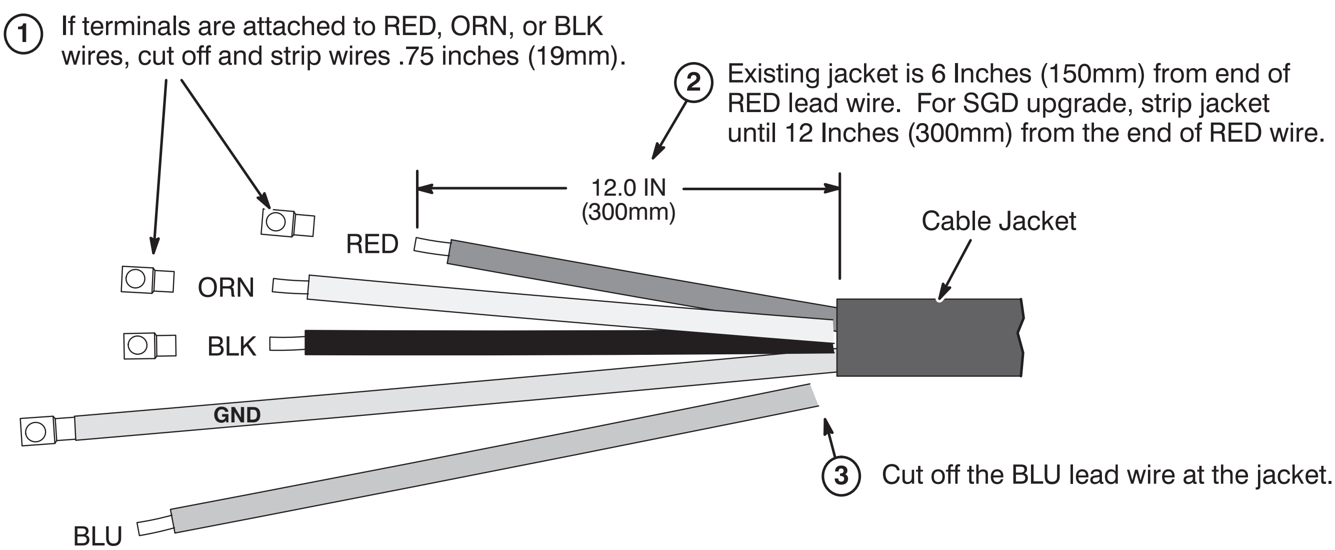
5. For Compact PDU, use furnished screwdriver (46–307819P125) to tighten terminal block screws!

2–7 MODIFY EXISTING RUN 042 FOR CONNECTION TO SGD GRADIENT CABINET

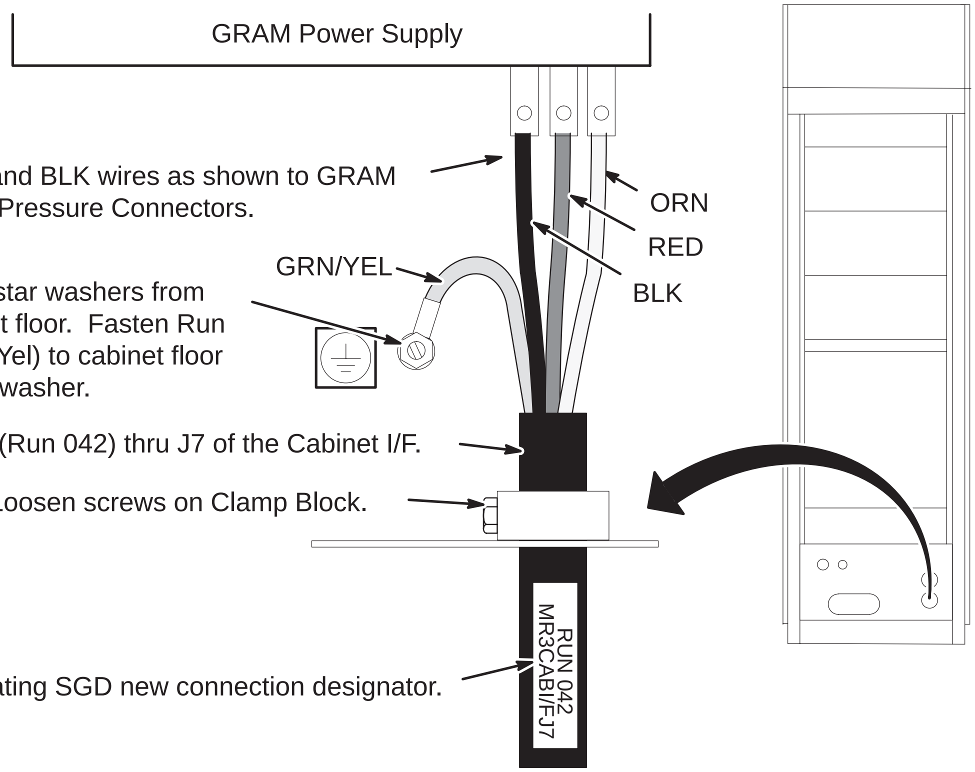
If the site being upgraded is a Horizon 8.x Base that had a 2–bay 8645 Gradient Cabinet and Run 042 is to be re–used, perform the following steps À, Á, and Â.



2–8 MODIFY EXISTING RUN 041 FOR CONNECTION TO SGD GRADIENT CABINET

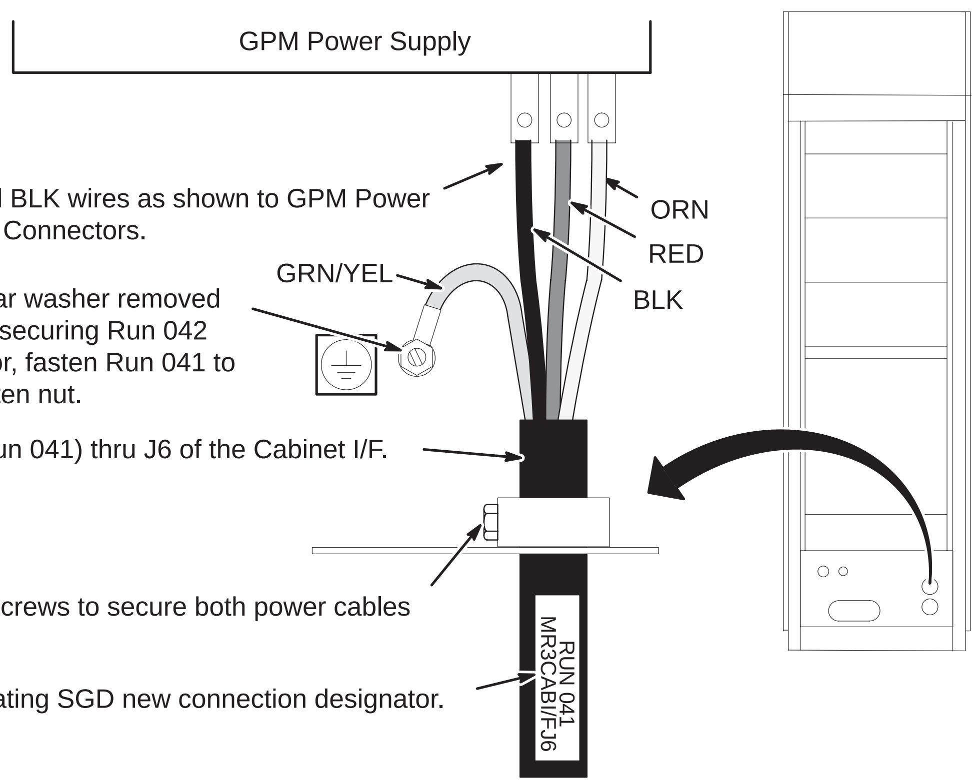


2–9 CONNECT RUN 042 TO SGD GRADIENT CABINET



4. Connect ORN, RED, and BLK wires as shown to GRAM Power Supply (lower) Pressure Connectors.
3. Remove (2) nuts and star washers from ground stud on cabinet floor. Fasten Run 042 ground wire (Grn/Yel) to cabinet floor using first nut and starwasher.
2. Route power cable (Run 042) thru J7 of the Cabinet I/F.
1. Loosen screws on Clamp Block.
5. Attach new label indicating SGD new connection designator.

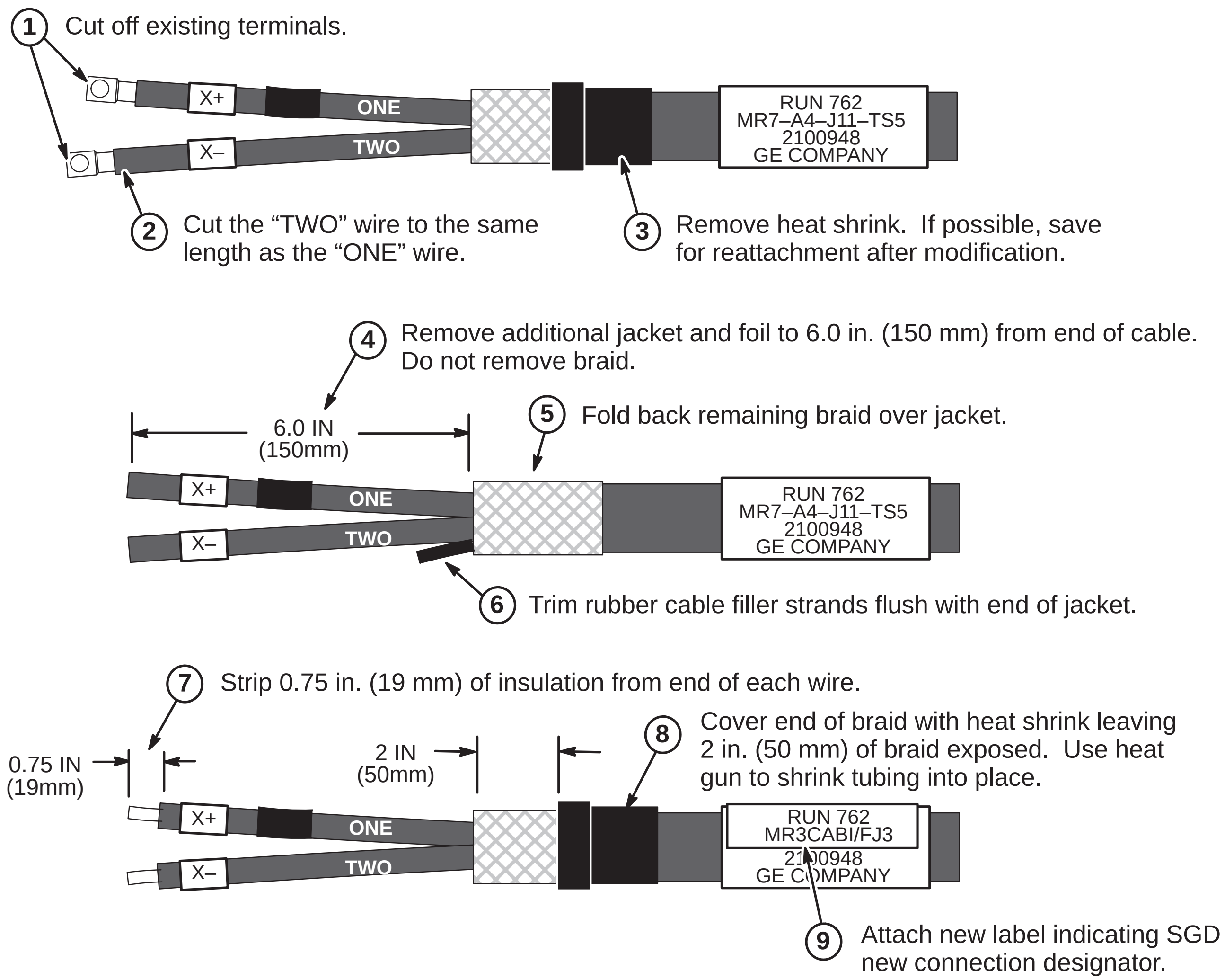
2–10 CONNECT RUN 041 TO SGD GRADIENT CABINET



3. Connect ORN, RED, and BLK wires as shown to GPM Power Supply (upper) Pressure Connectors.
2. Using second nut and star washer removed from ground stud before securing Run 042 GND wire on cabinet floor, fasten Run 041 to same ground stud. Tighten nut.
1. Route power cable (Run 041) thru J6 of the Cabinet I/F.
4. Tighten Clamp Block screws to secure both power cables to cabinet I/F.
5. Attach new label indicating SGD new connection designator.

2–11 MODIFY EXISTING RUNS 762, 763, AND 764 FOR SGD GRADIENT CABINET

The existing Output cable ends that are connected to the old 8645 Gradient or SGD Base Cabinet need to be modified for connection to the SGD Hi–Slew Gradient Cabinet.



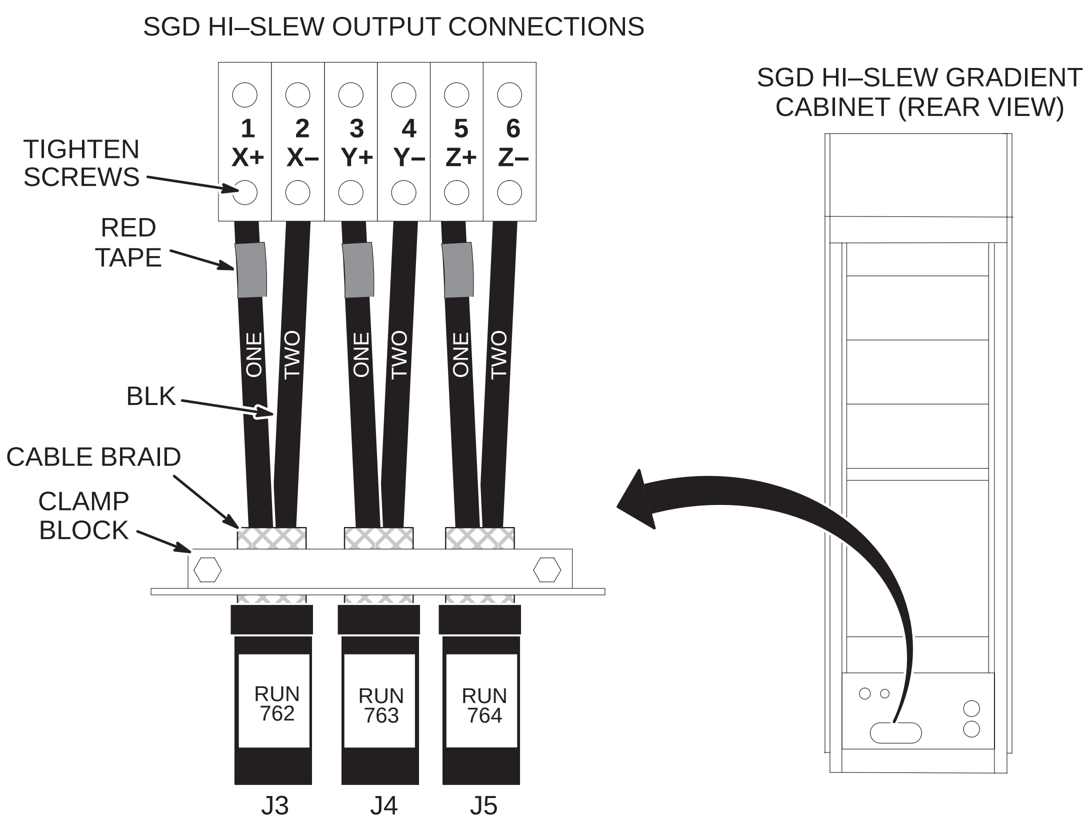
10 Repeat above process for Runs 763 and 764.
Note: Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**

2–12 CONNECT OUTPUT CABLE RUNS 762, 763, AND 764 TO SGD GRADIENT CABINET

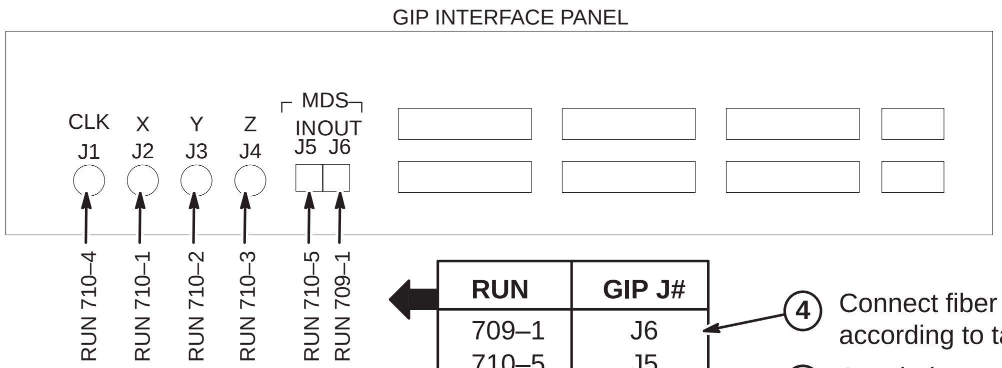
NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

1. Loosen screws on clamp block. Insert ends of gradient output cables into Cabinet Interface (Runs 762 into J3, Run 763 into J4, and Run 764 into J5) and position exposed braid in Clamp Block.
2. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

   NOTE: OVER TIGHTENING OF SCREWS IN STEP Â CAN DAMAGE TERMINAL BLOCK.
3. Tighten screws to <u>52–65 inch–lbs (5,88–7,34 N–M)</u> using supplied T–Wrench taped to bottom of cabinet. This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
4. Tighten clamp block to secure output power cables and ground the cable braid.



2–13 ROUTE AND CONNECT FIBER OPTICS TO GIP MODULE IN SGD GRADIENT CABINET



RUN	GIP J#
709–1	J6
710–5	J5
710–3	J4
710–2	J3
710–1	J2
710–4	J1

4 Connect fiber optics to GIP module according to table.

5 Attach the new labels below to each of the fiber optic runs connected in Step Ã.

RUN 709–1
MR3GIPJ6

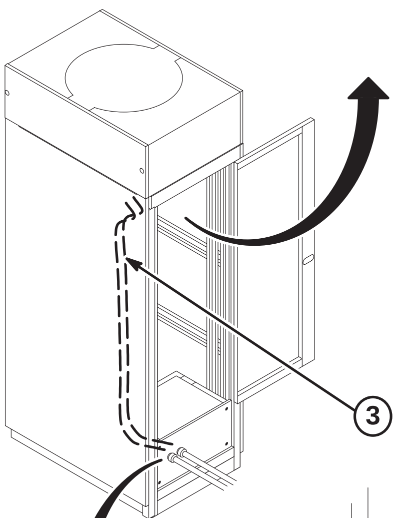
RUN 710–1
MR3GIPJ2

RUN 710–2
MR3GIPJ3

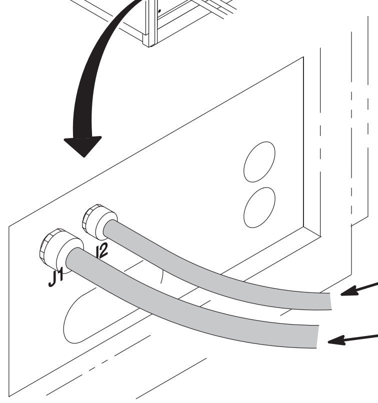
RUN 710–3
MR3GIPJ4

RUN 710–4
MR3GIPJ1

RUN 710–5
MR3GIPJ5



3 Route fiber optic cables to GIP module. Leave enough slack for extending GIP module forward.



2 Connect Run 709 to J2. Attach new label.

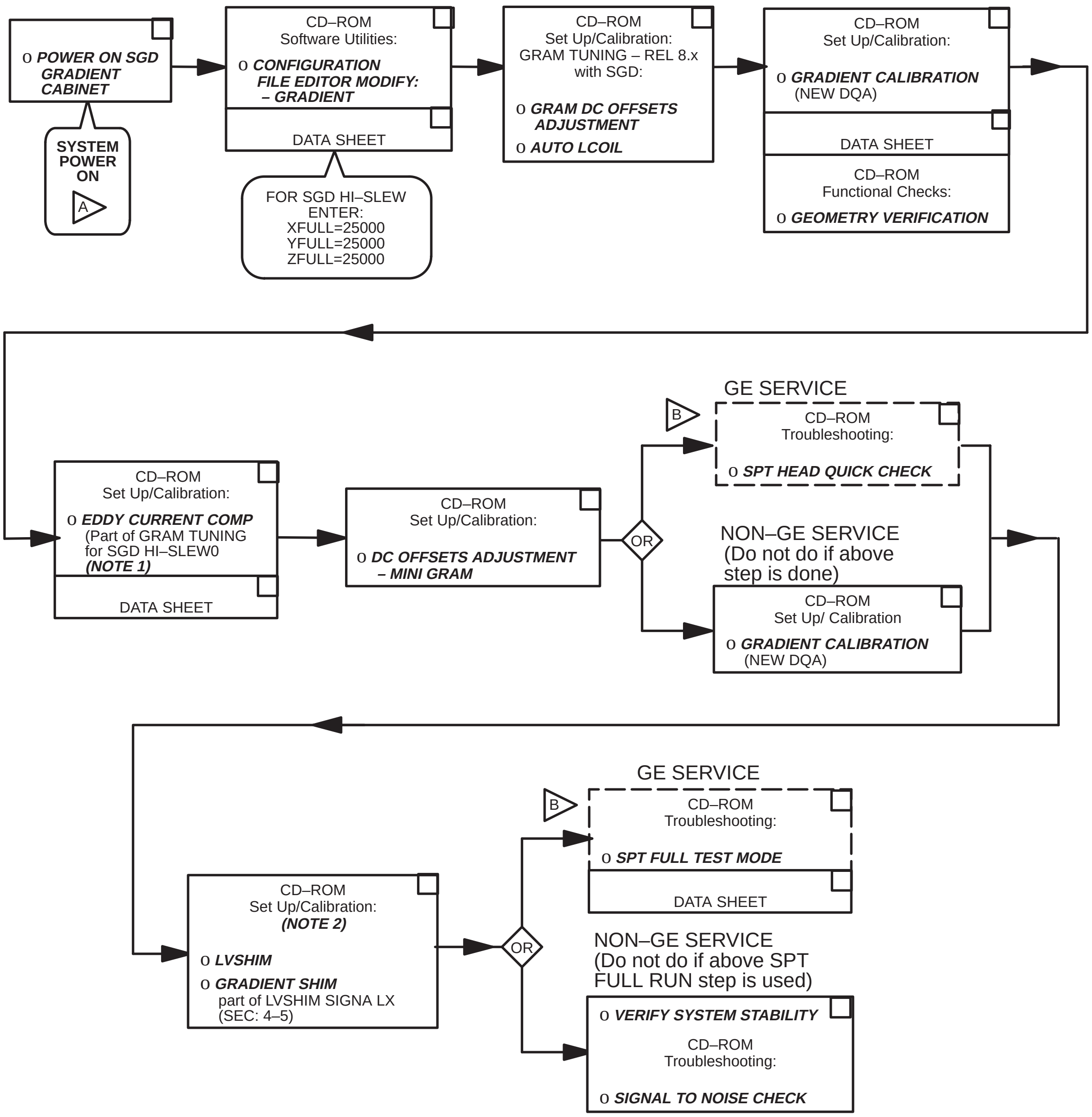
RUN 709
MR3CABI/FJ2

1 Connect Run 710 to J1. Attach new label.

RUN 710
MR3CABI/FJ1

SECTION 3 – CALIBRATION PROCESS

3–1 CALIBRATION FLOWCHART



(A) DO NOT TURN ON THE **SGD HI–SLEW** CABINET CIRCUIT BREAKERS (BREAKERS ON FRONT BOTTOM OF CABINET) UNTIL ROOM HUMIDITY IS AT 60% OR LESS.

CAUTION! EQUIPMENT IS SUSCEPTABLE TO FAILURE IN HIGH HUMIDITY.

(B) INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.

NOTE 1: VALUES OF EDDY CURRENT COMPENSATION ARE INDEPENDENT OF MAGNET SHIM.

NOTE 2: VALUES OF LVSHIM ARE DEPENDENT UPON EDDY CURRENT COMPENSATION.